

James Roggeveen, Ph.D.

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RESEARCH & TEACHING INTERESTS

My research program focuses on developing computational frameworks that integrate classical applied mathematics with modern machine learning. I aim to solve challenging inverse and optimization problems in fluid mechanics and soft matter, with a vision of creating differentiable pipelines to integrate physical models, data, and experiment. I am prepared to teach foundational and advanced courses in fluid mechanics, differential equations, and applied mathematics.

RESEARCH EXPERIENCE

2024 – **Postdoctoral Fellow**, Applied Mathematics, Harvard University, Cambridge, MA

Present *Faculty Advisor*: Michael P. Brenner

- Developed a novel meshless, differentiable spectral method for solving PDE inverse problems on complex geometries
- Developed a differentiable tool to automatically classify tensor-valued constitutive laws from sparse, one-dimensional experimental data, enabling robust material characterization from limited measurements.
- Co-led the creation of the HARDMath2 benchmark to evaluate the mathematical reasoning of large language models. Designed a novel collaborative framework and a custom LaTeX parser for automated, symbolic verification of AI-generated solutions.

2025 – **Visiting Researcher**, Google Deepmind, Cambridge, MA

Present – Developing evaluations for state-of-the-art large language models.

EDUCATION

2024 **Doctor of Philosophy** in Mechanical Engineering, Princeton University

Faculty Advisor: Howard A. Stone

Thesis: “Asymmetric and rough boundary surface effects on fluid flows”

2019 **Master of Advanced Study** in Applied Mathematics, University of Cambridge

Result: Passed with Distinction

2018 **Bachelor of Science** in Mechanical Engineering, Massachusetts Institute of Technology

GPA: 5.0

PUBLICATIONS

* denotes equal contribution

Preprints

J. V. Roggeveen and M. P. Brenner, Meshless solutions of PDE inverse problems on irregular geometries. arXiv:2510.25752

A. M. Sunol*, **J. V. Roggeveen***, M. G. Alhashim, H. S. Bae, M. P. Brenner, Learning constitutive models and rheology from partial flow measurements using automatic differentiation. arXiv:2510.24673.

Manuscripts under review

H. Pan, **J. V. Roggeveen**, *et. al.*, CMT-Benchmark: A Benchmark for Condensed Matter Theory Built by Expert Researchers. ICLR, under review. arXiv:2510.05228

Peer-reviewed publications

1. **J. V. Roggeveen***, E. Wang*, *et. al.*, HARDMath2: A Benchmark for Applied Mathematics Built by Students as Part of a Graduate Class. *NeurIPS*, 2025. arXiv:2505.11774
2. H. H. Cheng, **J. V. Roggeveen**, H. Wang, H. A. Stone, Z. Shi, C. P. Brangwynne, Micropipette aspiration reveals differential RNA-dependent viscoelasticity of nucleolar subcompartments. *Proceedings of the National Academy of Sciences*, 2025. DOI:10.1073/pnas.2407423122
3. **J. V. Roggeveen** and H. A. Stone, Drift of elastic hinges in quasi-two-dimensional oscillating shear flows. *Physical Review Fluids*, 2025. DOI:10.1103/PhysRevFluids.10.034401
4. **J. V. Roggeveen**, H. Wang, Z. Shi and H. A. Stone, A calibration-free model of micropipette aspiration for measuring properties of protein condensates. *Biophysical Journal*, 2023. DOI:10.1016/j.bpj.2023.09.018
5. **J. V. Roggeveen**, H. A. Stone and C. Kurzthaler, Transport of a passive scalar in wide channels with surface topography: An asymptotic theory. *Journal of Physics: Condensed Matter*, 2023. DOI:10.1088/1361-648X/acc8ad
6. **J. V. Roggeveen** and H. A. Stone, Motion of asymmetric bodies in two-dimensional shear flow. *Journal of Fluid Mechanics*, 2022. DOI:10.1017/jfm.2022.203
7. **J. V. Roggeveen**, A. Stoica and M. Dolci, Analysis of alternative energy harvesting methods to power atmospheric robotic explorers on Jupiter. In *proc. of IEEE Aerospace Conf.*, 2017. DOI:10.1109/AERO.2017.7943715

PRESENTATIONS

Invited talks

- J. V. Roggeveen, Beyond the Rheometer: Integrative Approaches to Modeling and Characterizing Complex Fluids, *Invited Seminar*, Nanyang Technological University, Singapore, 2025.
- J. V. Roggeveen, Geometry-driven drift of passive particles in flows, *Soft, Fluid, Living Matter Coffee Hour*, Yale University, New Haven, 2025.
- J. V. Roggeveen, Geometry-driven drift of passive particles in flows, *Physical Mathematics Seminar*, MIT, Cambridge, 2025.
- J. V. Roggeveen, Geometry-driven drift of passive particles in flows, *Modeling and Simulation Group Meeting*, NYU Courant Institute, New York, 2025.
- J. V. Roggeveen, Shear-driven drift of passive, asymmetric and deformable particles, *Mechanics Joint Group Meeting*, Boston University, Boston, 2025.
- J. V. Roggeveen, Multiphase flows at low Reynolds numbers: From condensate rheology to fluid-structure interactions, *Center for Computational Biology*, Flatiron Institute, New York, 2023.
- J. V. Roggeveen, Measuring condensate rheology with micropipette aspiration & fluid-structure interactions at low Reynolds number, *LadHyX Seminar*, École polytechnique, Paris, France, 2023.
- J. V. Roggeveen, H. Wang, Z. Shi and H. A. Stone, Material characterization of condensates using micropipette aspiration, *Phase Group Seminar*, Princeton, 2022.
- J.V. Roggeveen and H. A. Stone, Motion of asymmetric bodies in shear flow, *Soft Materials Coffee Hour*, Princeton, 2021.

Conferences

- J. V. Roggeveen, A. Sunol, M. Alhashim, M. P. Brenner, Fitting tensorial constitutive models to experimental rheology using automatic differentiation, *APS Global Physics Summit*, Anaheim, 2025.

J. V. Roggeveen and H. A. Stone, Drift of elastic hinges in oscillating shear flows, *77th Annual Meeting of the APS Division of Fluid Dynamics*, Salt Lake City, 2024.

J. V. Roggeveen and H. A. Stone, Purcell's elastic swimmer: Drift of elastic hinges in oscillatory shear flows, *76th Meeting of the APS Division of Fluid Dynamics*, Washington D.C., 2023.

J. V. Roggeveen, H. Wang, Z. Shi and H. A. Stone, Modeling micropipette aspiration for use in material characterization of biological condensates, *XIXth International Congress on Rheology*, Athens, Greece, 2023.

J. V. Roggeveen, H. A. Stone and C. Kurzthaler, Transport of a passive scalar in wide channels with surface topography, *75th Meeting of the APS Division of Fluid Dynamics*, Indianapolis, 2022.

J.V. Roggeveen and H. A. Stone, Motion of asymmetric bodies in shear flow, *74th Meeting of the APS Division of Fluid Dynamics*, Phoenix, 2021.

TEACHING EXPERIENCE

Lecturer

Served as sole instructor for a graduate-level applied mathematics course, designing all lectures, problem sets, and exams. Pioneered a novel pedagogical approach by leading the class in a collaborative project to create, peer-review, and test problems for AI evaluation.

- AM 201: Physical Mathematics I, Fall 2025, Harvard University

Teaching assistant

Responsibilities included: developing course materials including problem sets and exams; teaching precept sections, grading assignments, projects, and exams; providing individualized help to students in office hours; lab supervision and training

- AM 201: Physical Mathematics I, Spring 2025, Harvard University
- Mathematics in Engineering I, Fall 2023, Princeton University
- Viscous Flows and Boundary Layers, Spring 2023, Princeton University
- Mathematical Methods of Engineering Analysis I, Fall 2022, Princeton University
- Instabilities in Fluids: Linear and Non-Linear Analysis of Waves and Patterns in the Environment, Spring 2022, Princeton University
- Multivariable Calculus, Fall 2021, Princeton University
- Thermodynamics, Fall 2017, MIT
- Thermal-Fluids Engineering I, Spring 2017, MIT
- Design and Manufacturing I, Undergraduate Assistant, Spring 2017, MIT

Additional teaching experience/Outreach

- MIT Global Teaching Labs South Korea, Instructor, Spring 2018
- 7.012 Introductory Biology, Tutor, Fall 2015
- China Education and Technology Initiative, Instructor, Summer 2015

FELLOWSHIPS & AWARDS

Fellowships

Guggenheim Second Year Fellowship, Princeton University, 2020

Teaching awards

Graduate School Teaching Award, Princeton University, 2022

Academic recognition

Sayre Award for Academic Excellence, Princeton University, 2020

PROFESSIONAL SERVICE

Leadership

Chair, Graduate Student Council, Princeton MAE, 2019 – 2024

Captain, MIT Asian Dance Team, 2014 – 2018

Scholarly Service

Session Chair, “Fluids I”, APS Global Physics Summit, 2025

Reviewer, *Physical Review Fluids*, American Physical Society

Senior Reviewer, MIT Engineering Advisory Board 2019 – 2023

Miscellaneous

Resident Graduate Student, Mathey College, Princeton University, 2021 – 2022

REFERENCES

Michael P. Brenner, brenner@seas.harvard.edu

Howard A. Stone, hastone@princeton.edu

Luc Deike, ldeike@princeton.edu

Pierre-Thomas Brun, pierre-thomas.brun@kuleuven.be